

Liver Radiodensity and Chemotherapy Toxicity

Hospital-Level Variation in a Multi-centre Cohort

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COLON Study · Statistical Consulting Project

The clinical question

Diagnosis



Toxicity

Liver HU

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is it associated with the timing of toxicity?

Capecitabine and oxaliplatin cause toxicities that force dose reductions, delays, or stopping treatment.
Liver fat at diagnosis is a plausible candidate marker.

The data

Cohort: COLON study, prospective, 11 Dutch hospitals.

Three endpoints, separate models:

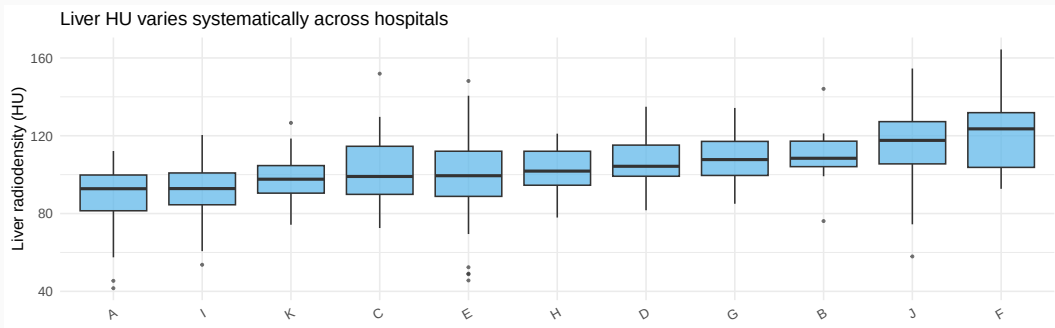
- CAP: capecitabine toxicity
- OXA: oxaliplatin toxicity
- CAPOX: combined regimen toxicity

Time: chemotherapy cycles to first dose-limiting toxicity (or censored at last cycle).

Adjusted for: age, sex, BMI, smoking, treatment-specific starting dose.

Endpoint	N	Events	Event rate
CAP	314	164	52%
OXA	289	236	82%
CAPOX	305	234	77%

What makes this tricky



11 hospitals means 11 different CT scanners. Liver HU differs significantly between centres (ANOVA, $p < 0.001$), most likely because of scanner calibration. Toxicity rates also differ for OXA, but for CAP and CAPOX the variation is consistent with sampling noise.

So hospital looks like a confounder we should not just ignore.

How we handle it

Unstratified Cox

$$h(t \mid \mathbf{x}) = h_0(t) e^{\beta \text{HU} + \gamma^\top \mathbf{x}}$$

One baseline hazard for everyone. The estimate of β mixes within-hospital and between-hospital differences in liver HU.

Stratified by hospital

$$h_k(t \mid \mathbf{x}) = h_{0k}(t) e^{\beta \text{HU} + \gamma^\top \mathbf{x}}$$

Each hospital k has its own baseline. The estimate of β comes from *within-hospital* comparisons only.

What we did

Fit both specifications for each endpoint and compare. The difference between the two estimates tells us how much the unstratified version is being affected by between-hospital differences.

The main result

CAP and OXA

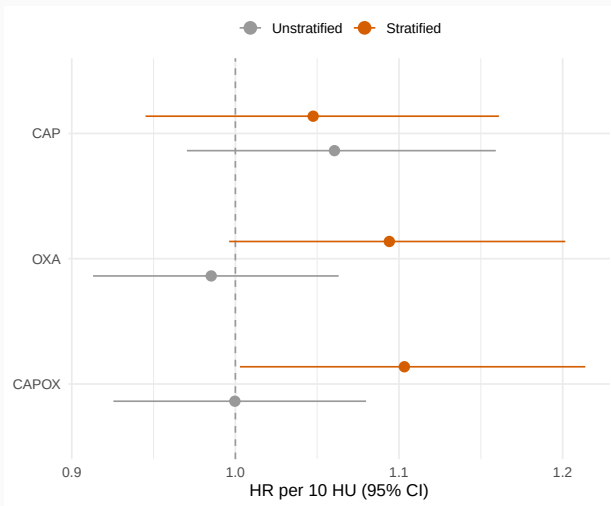
Neither specification reaches statistical significance.

CAPOX

Unstratified: $HR \approx 1.00$, no association.

Stratified: $HR = 1.10$ (1.00 – 1.21), $p = 0.044$.

For CAPOX, between-hospital differences in measured HU were hiding the within-hospital signal.



How big is the effect, in context?

For CAPOX (the conventionally significant endpoint):

Scale	HR	95% CI
per 10 HU	1.10	1.00 – 1.21
per 1 SD ($\approx$ 19 HU)	1.21	1.01 – 1.45

A patient with a fatty liver compared to one with a healthy liver, about one within-cohort SD apart, has roughly a **21% higher hazard** of CAPOX toxicity.

Caveats

The lower bound of the CI is right at 1, the cohort is observational, and we have not validated this in another dataset. The effect is real in this cohort, but it is not yet a clinical tool.

Takeaways

1. **Stratification really mattered for our results.**

Without it we would have concluded *no association* for CAPOX. Hospital-level CT calibration was hiding the within-hospital signal.

2. **There is a small within-hospital effect for CAPOX.**

HR $\approx$ 1.10 per 10 HU, or about 1.21 per SD. No effect for CAP or OXA.

3. **Not a clinical tool yet.**

Effect is modest, the CI is wide, and the design is observational. The transferable lesson is methodological: in multi-centre cohorts with measurement heterogeneity, the within-hospital contrast can look quite different from the pooled one.

Backup material

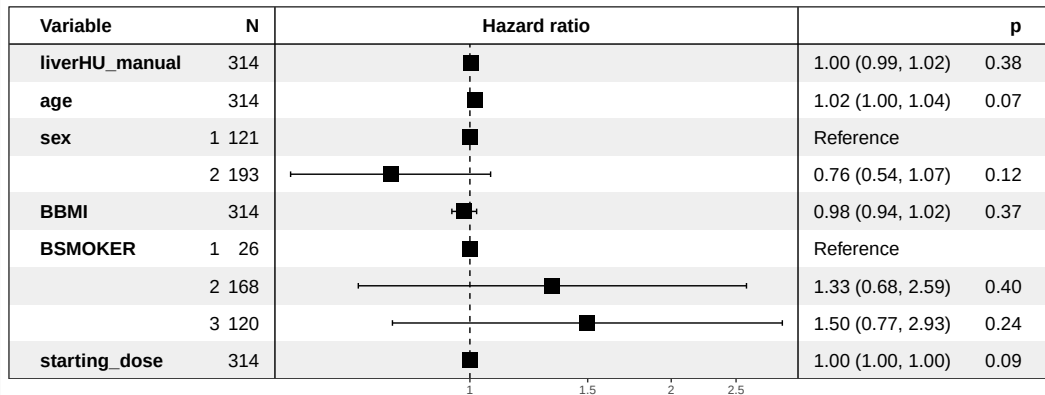
Hospital toxicity rates: chi-square test

Tests whether observed event-rate differences across hospitals exceed binomial sampling noise around a common rate.

Endpoint	N	Chi-sq	df	p-value
CAP	314	16.90	10	0.077
OXA	289	30.86	10	<0.001
CAPOX	305	15.16	10	0.126

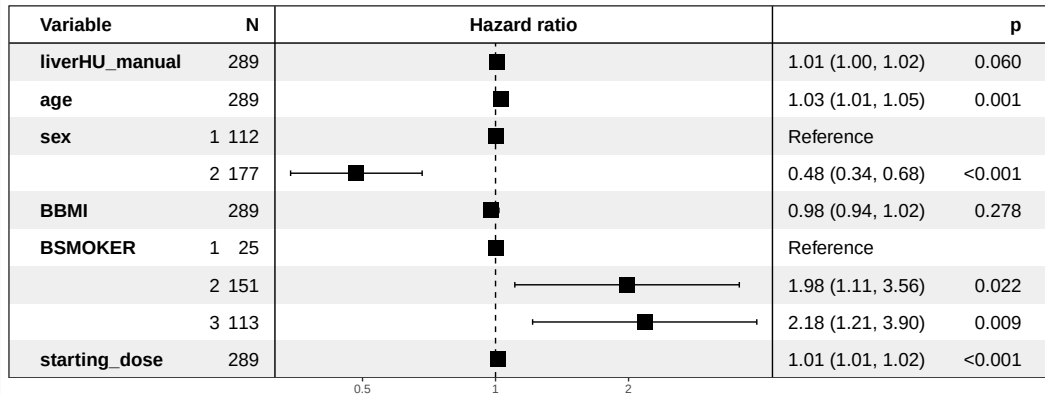
- Smallest hospital $\times$ event cell count is 7, so the asymptotic chi-square is reliable (no need for Monte Carlo simulation).
- For OXA the differences are real. For CAP and CAPOX the observed range is consistent with sampling noise around a common rate.
- Stratification is still justified by the liver HU heterogeneity (ANOVA), regardless

Full covariate model: CAP



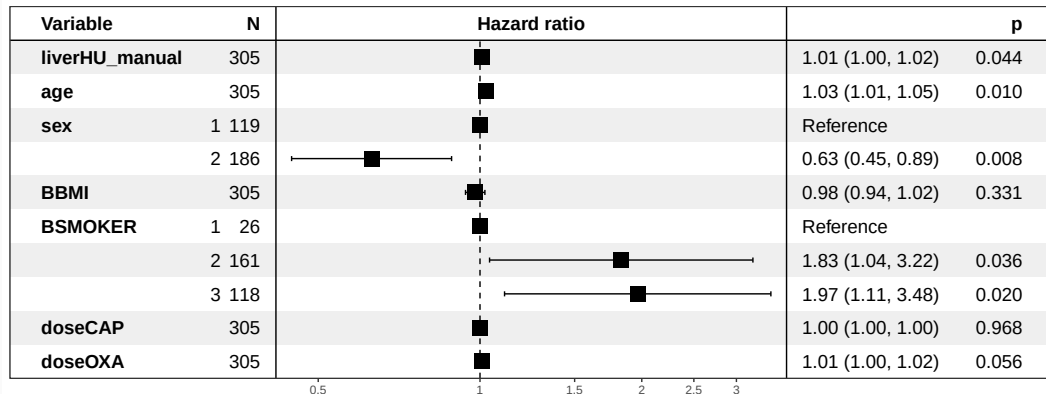
For CAP, none of the covariates reach statistical significance. The CAP model is generally less informative, suggesting that capecitabine-related toxicity is less strongly tied to the patient characteristics included here.

Full covariate model: OXA



For OXA, age, sex (male less affected), smoking status, and oxaliplatin starting dose are all significantly associated with toxicity. Liver HU sits just outside conventional significance ($p = 0.060$).

Full covariate model: CAPOX



For CAPOX, age, sex, and smoking show consistent associations with toxicity, similar to the OXA model. Liver HU reaches conventional significance ($p = 0.044$).

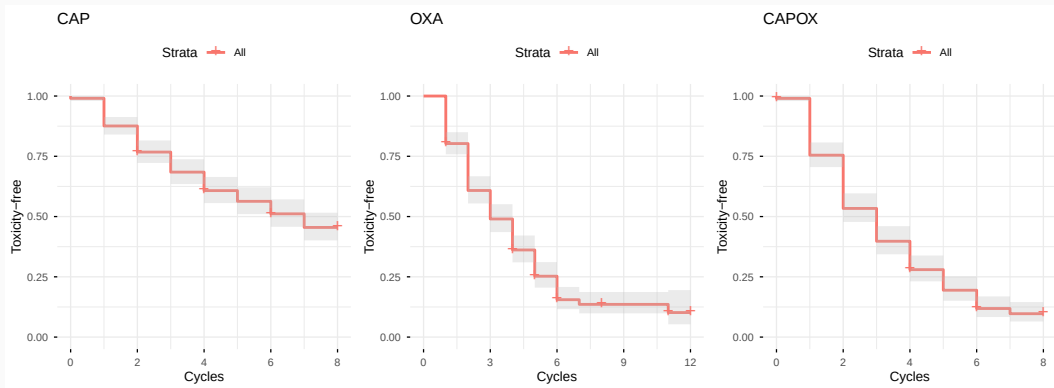
Proportional hazards check

Schoenfeld residual test, GLOBAL row per model. A non-significant p means no evidence the proportional-hazards assumption is violated.

Endpoint	Chi-sq	df	p-value
CAP	5.54	7	0.594
OXA	9.33	7	0.230
CAPOX	3.94	8	0.863

All three GLOBAL p-values are well above 0.05. Per-covariate Schoenfeld results (also non-significant) are in the analyst notebook.

Kaplan-Meier curves



Most events occur in the first few cycles. The probability of remaining toxicity-free drops sharply early on, then stabilises as fewer patients remain at risk.

Sample sizes per hospital (CAP)

Hospital	N	Events	Rate (%)
A	70	37	53
E	58	34	59
J	43	25	58
C	35	16	46
I	30	10	33
K	25	16	64
H	23	8	35
D	18	9	50
F	16	10	62
G	16	8	50
B	10	8	80

Hospital sizes range from very small to ~80 patients. The smallest hospitals contribute little to the within-hospital comparisons, which is appropriate: stratification gives them their own baseline rather than letting them drag a pooled estimate around.